Math 556: Graph Theory
Homework

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§1.1

1. Consider the following four families of graphs:

   $$A = \{\text{paths}\}, \qquad B = \{\text{cycles}\}, \qquad C = \{\text{complete graphs}\}, \qquad D = \{\text{bipartite graphs}\}.$$

   For each pair of these families, determine all isomorphism classes of graphs that belong to both families.

2. Prove that removing two opposite corner squares from an 8-by-8 checkerboard leaves a subboard that cannot be partitioned into 1-by-2 and 2-by-1 rectangles. Using the same argument, make a general statement about all bipartite graphs.

   Hint: An 8-by-8 checkerboard can be represented by a graph. Each square corresponds to a vertex, and two vertices are adjacent if and only if the corresponding squares share a side.

3. Let G be a simple bipartite graph with n vertices and m edges. Prove that

   $$m \leq \frac{n^2}{4}.$$

   Hint: Suppose X and Y are partite sets of G with $|X| = r$ and $|Y| = s$. First show that

   $$m \leq rs,$$

   and use it to prove

   $$m \leq \frac{n^2}{4}.$$

4. Let G be a graph with girth 4 in which every vertex has degree k. Prove that G has at least $2k$ vertices. Determine all such graphs with exactly $2k$ vertices.

5. Let P be the Petersen graph. Use the definition of P given in terms of 2-element subsets of a set of size 5 to prove the following.

(a) P contains a cycle of length 6.

(b) P contains a cycle of length 8.

(c) P does not contain a cycle of length 7.

Hint: For contradiction, assume that C is a cycle in P of length 7. First, explain why every edge of P whose endpoints both lie in C must be an edge of C.

Let u be a vertex of C and let v, w be vertices of C farthest from u. Show that there is some vertex

$$x \notin V(C)$$

such that u and v are both adjacent to x, and u and w are both adjacent to x. Conclude that $\{w, v, x\}$ are the vertices of a 3-cycle in P, which contradicts the fact that P is triangle-free.